GlandersWatch: A Modular Platform for Glanders Risk Prediction, Retrieval-Augmented Explanation and Owner Alerting

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Certificate

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Dr. Sangita Dhara
(Supervisor)

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Abstract

Glanders is a bacterial zoonosis of equids caused by *Burkholderia mallei*. It is notifiable, frequently fatal, and remains endemic in parts of Africa, Asia and South America. Indian surveillance between 2019 and 2023 identified 728 seropositive equids across 138 districts, with strong geographic concentration. The published machine learning literature on this disease is confined to laboratory identification of the organism by spectroscopic methods, and no population-level geographic risk model has been reported.

This report presents GlandersWatch, a platform built from three independently deployable components: a calibrated machine learning risk engine, a retrieval-augmented layer that explains each prediction with citations to official guidance, and an alert service that delivers the result to equine owners. The decomposition is the contribution as much as the model is. Each component communicates through a versioned schema rather than shared internals, none hardcodes the disease, and each degrades without failing the others.

The risk engine combines four public datasets not previously applied to glanders into a country-year panel: a text-mined occurrence corpus from Europe PMC, FAOSTAT livestock stocks, World Bank development indicators and NASA POWER reanalysis climatology. 6,700 country-year rows reduce to 2,880 modelling observations across 160 equine-holding countries from 2005 to 2023, with 169 positive labels. Candidate features are screened by attribution rather than association: the top 8 of 42 numeric candidates, ranked by mean absolute SHAP on the training years alone, are kept, with no categorical columns, and six model families are trained under three validation schemes.

Two tests were run, because they answer different questions. The first asks how well the model predicts years it has never seen: the best model reaches a precision-recall AUC of 0.495 against a base rate of 5.87%, roughly eight times better than chance, which is useful but not accurate enough to act on without a person in the loop. The second asks how well it predicts countries it has never seen, which is what deployment actually requires. That test changes the ranking. Logistic Regression leads at 0.381 and its probabilities are well calibrated, with an error of 0.025, so a predicted 0.20 means roughly one chance in five. It is therefore the champion and the estimate to quote for an unseen country.

An easier question was tested alongside. Instead of asking whether glanders appears in one particular year, ask whether a country reports glanders at all. Discrimination there rises to about 0.95. That is not a better model but a simpler question: whether a country reports the disease is a stable national characteristic, while the timing is the part that is genuinely hard.

The main limitation follows from the label. A positive means glanders was documented, not that it occurred, and a negative may mean no disease or simply no report. The model therefore learns where glanders is reported, and reporting effort is the principal confounder: a country with a reference laboratory and an open-access publishing record looks riskier than a country with the same disease burden and weaker surveillance. Estimating and correcting for that capacity is the main item of future work.

The explanation and alert components are described through their interfaces, rules and evaluation plan. They carry no measured results because they are not deployed, and measuring them is what a pilot would supply.

Keywords: Glanders, *Burkholderia mallei*, Risk Prediction, Gradient Boosting, Probability Calibration, SHAP, Retrieval-Augmented Generation, Citation Grounding, Alert Fatigue, Modular System Design, MLOps, Animal Health Surveillance.

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Chapter 1

Introduction

1. Background

Glanders is a contagious and often fatal disease of horses, donkeys and mules caused by *Burkholderia mallei*. The organism has no environmental reservoir, and asymptomatic carriers remain infectious for long periods. Human infection is rare but carries a high case fatality rate and follows occupational contact. The disease is notifiable to the World Organisation for Animal Health and is classified as a potential bioterrorism agent.

The economic weight sits in low and middle income countries, where equids remain essential to transport, construction and agriculture. An equid is a member of the family *Equidae*: horses, donkeys (recorded here as asses) and mules. India held 5.50 lakh of them in the 2019 Livestock Census, made up of 3.42 lakh horses, 1.24 lakh asses and 0.84 lakh mules. That figure is the size of the susceptible population, not a count of sick animals: it says how many animals are exposed, not how many the disease has damaged. Most are held by households that depend on them for daily income, so the death of one working animal is a direct loss of livelihood.

The damage that is documented is smaller and better measured. A national programme screening 158,798 sera from 391 districts between 2019 and 2023 found 728 seropositive animals in 138 districts, roughly sixty per cent of them in Uttar Pradesh; earlier surveillance covering 2006 to 2018 reported 1,145 positives among 128,404 sera. That is under one per cent of the equid population, concentrated in a few states, and because glanders is notifiable and frequently fatal once clinical, each positive is acted on rather than monitored.

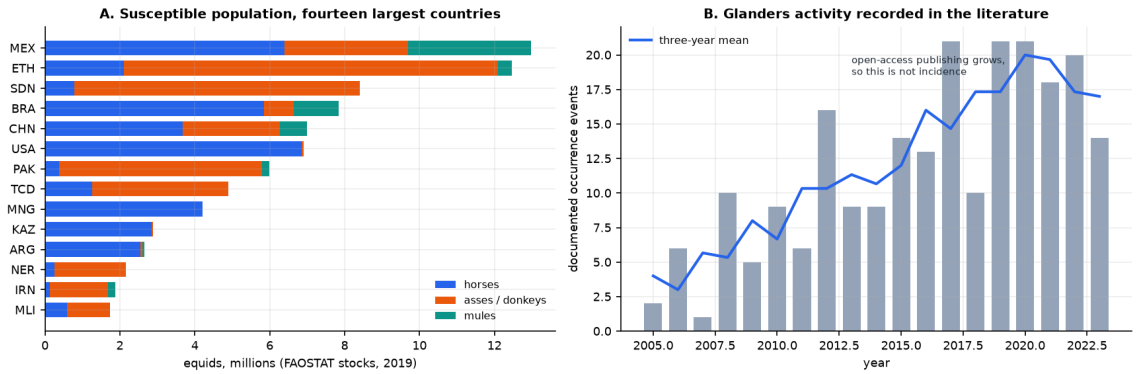


Figure 1.1: The two quantities behind the problem. Panel A gives the equid population of the fourteen largest countries, split by species, which sets the size of the exposed population. Panel B gives the occurrence events recovered from the literature each year, which sets the size of the label. The rise in panel B follows open-access publishing volume as much as disease, which is why the label is treated as documented activity throughout.

2. Problem Definition

A district veterinary officer faces a narrow question: is glanders likely to appear in this area in the coming year? Answering it means combining the history of the disease nearby, the size of the susceptible population, and the conditions that permit transmission.

Surveillance reports what has already happened, and forecasting systems exist for several livestock diseases but not for glanders. A prediction alone would still not be usable: an owner cannot act on a probability of 0.24, and an officer cannot defend a forecast whose basis is hidden. A warning that arrives for every area every month also teaches its recipients to ignore it.

The problem therefore has four parts. Two concern producing the prediction, two concern turning it into something a person can act on.

1. **Data and label.** No public source records glanders occurrence completely, so occurrence must be reconstructed from the literature and joined to population and environmental covariates at a common resolution. The outcome is then rare, so accuracy is useless and a negative observation needs an explicit meaning.
2. **Model selection under distribution shift.** A model fitted on some countries must be evaluated on countries it has never seen, because that is the deployment condition.
3. **Grounded explanation.** The prediction must be explained in words a non-specialist can act on, every claim must trace to official control guidance, and the layer must refuse rather than generate an answer the corpus does not support.
4. **Delivery without alert fatigue.** A probability is not an alert. The operating point and the message rules must keep volume low enough that messages are read, while the threshold stays defensible against the events caught.

3. Objectives

The project has four objectives.

1. **The prediction study.** Assemble a country-year panel from public sources not previously applied to the disease, compare six model families under three validation schemes, and report the achievable skill including negative results.
2. **Prediction as a service.** Package the trained model as a versioned artefact behind a REST interface, so any consumer can call it without knowing how it works.
3. **Cited explanation.** Build a retrieval-augmented layer that explains a prediction from official guidance, cites every claim, and refuses rather than guesses.
4. **Owner alerts.** Convert a calibrated probability into few enough messages that they are read, using threshold, suppression and deduplication rules.

All four share one requirement: each component must be independently testable and replaceable, and free of any hardcoded reference to glanders.

4. Scope and Limitations

The study is scoped to country-year resolution. District-level modelling for India was envisaged but is not attempted, because the covariates could not be assembled from public sources within the project period. The explanation and alert layers are resolution-agnostic and apply unchanged once a district-level model exists. The model is a population-level tool: it does not diagnose an individual animal and does not replace laboratory confirmation.

Three limitations shape every result. The label comes from published reports, so a negative may mean no disease or no reporting. Occurrence is reconstructed by an extractor not validated against a hand-annotated set. And with 169 positives the uncertainty on every headline metric is substantial, which is quantified rather than hidden.

Chapter 2

Literature Review and Gap Analysis

1. Epidemiology and Surveillance

Glanders is treated as a neglected zoonosis with high case fatality and a real occupational risk to handlers [1, 2]. A recent systematic review and meta-analysis consolidates seroprevalence estimates and identifies management practice and animal movement as the dominant drivers [3]; regional reviews document continued circulation in Brazil [4]. The disease persists where equid populations are large, animals move often and surveillance is weak. The first two are measurable from public data, while the third decides whether the disease is visible at all and is therefore a confounder rather than a covariate.

Indian surveillance is the strongest evidence base. Malik et al. described outbreaks reported between 2006 and 2011 [7], Singha et al. reported serological surveillance for 2015 to 2018 [6], and the national programme covering 2019 to 2023 produced a prevalence estimate with a risk analysis [5]. These establish where glanders occurs and which animal-level factors the risk analysis identified. None builds a predictive model, and none uses population or reanalysis covariates as predictors.

2. Existing Forecasting Systems in India

The closest operational system is NADRES v2, the National Animal Disease Referral Expert System run by ICAR-NIVEDI [39]. It combines outbreak records from AICRP centres with meteorological and remote-sensing covariates and issues district-level forewarnings about two months ahead, using a model family that overlaps with the one used here.

This project differs in four complementary ways. It is glanders-specific, where a national system must spread effort across many diseases. It uses only public data, so anyone can reproduce and audit it. It attaches a cited explanation to each prediction, which matters when the reader is an owner rather than a trained analyst. And it delivers to the owner directly, with rules that suppress routine noise. NADRES has institutional access this project lacks; this project has a level of local explainability that a multi-disease platform does not attempt.

3. Spatial Risk and Machine Learning

Spatial work on glanders is descriptive. Fonseca-Rodríguez et al. applied spatiotemporal cluster detection to Brazilian data [8], and species distribution models have been compared as tools for livestock disease [9]. Cluster detection identifies where cases have already concentrated; it does not forecast which cells will be positive next period, which is the question an early warning system must answer.

Predictive modelling is established for avian influenza. Kane et al. compared ARIMA and random forest models for H5N1 outbreaks [11], Walsh et al. used machine learning to target active surveillance [12], and Yoo et al. and Alshatti et al. built farm-level risk models in Korea and Kuwait [13, 10]. That literature supplies the methodological template used here: assemble occurrence, population and environmental covariates at a common unit, fit tree-based ensembles, and evaluate on held-out periods. The template has been applied to influenza, not to glanders.

The only machine learning work specific to *B. mallei* is spectroscopic identification. Moawad et al. built a Raman assay in which learned models classify spectra [14], and Stöckel et al. reported spectroscopic detection of *B. mallei* and *B. pseudomallei* [15]. Both are laboratory diagnostic problems; neither addresses geographic or temporal distribution, and neither produces a population risk estimate. Using published literature as structured occurrence data has precedent in epidemic intelligence [16, 17], but here event extraction is the primary source of the label rather than a supplementary signal, because official outbreak records are incomplete for many countries. The label therefore inherits the reporting biases of the literature, which is analysed rather than assumed away.

4. Prior Art Audit

To test the novelty claim rather than assert it, a reproducible set of queries was run against Europe PMC. Each row of Table 2.1 is a query count; the query strings and retrieved titles are retained with the project code.

Table 2.1: Europe PMC hit counts for combinations of glanders with each data source and method used in this study.

Query	**Hits**
glanders (baseline corpus)	6674
glanders + machine learning	75
glanders + tree ensembles	40
glanders + forecasting	139
glanders + FAOSTAT	7
glanders + NASA POWER	0
glanders + World Bank indicators	22
glanders + equine population denominator	54
glanders + Europe PMC text mining	378
glanders + chained/stacked models	36
glanders + random forest (risk context)	8
glanders + niche / distribution model	62
glanders + remote sensing / reanalysis	10
glanders + spatio-temporal clustering	463
glanders + gradient boosting	4

The decisive cells pair glanders with NASA POWER, which returns zero records, and with gradient boosting, which returns four, none a glanders risk model.

5. What the Existing Work Leaves Open

The evidence gap. The literature divides into three disjoint groups. Epidemiological studies describe where glanders occurs; machine learning studies of *B. mallei* classify laboratory spectra; veterinary risk models exist but are built for influenza. No study joins an occurrence corpus, a livestock census, development indicators and reanalysis climatology into a single panel for glanders, and none evaluates a calibrated predictive model on that panel.

Deployment gap. A probability on a dashboard does not reach the person who owns the animals, and a risk score without a stated basis is not something a field officer will act on. The literature stops short of the delivery problem: how a forecast becomes a cited explanation, and how that becomes a message an owner trusts.

The contribution has three parts: a calibrated risk model, evaluated under forward temporal and country-holdout splits and served as a versioned artefact; a retrieval-augmented layer that grounds each prediction in official control guidance, cites every claim and refuses rather than generates; and an alert service whose threshold, suppression and deduplication rules turn a probability into a small number of messages. All three exchange fixed versioned contracts and none hardcodes the disease.

Chapter 3

System Architecture

1. What the Platform Has to Do

The platform answers one question: whether glanders is likely to be documented in an area next period. It produces three outputs from it, a number on a map, an explanation in words and a message to an owner, each read by a different person on a different device, and each able to fail without the others failing. One service serving all three would tie alert delivery to the latency of an LLM call and make the risk model impossible to test alone.

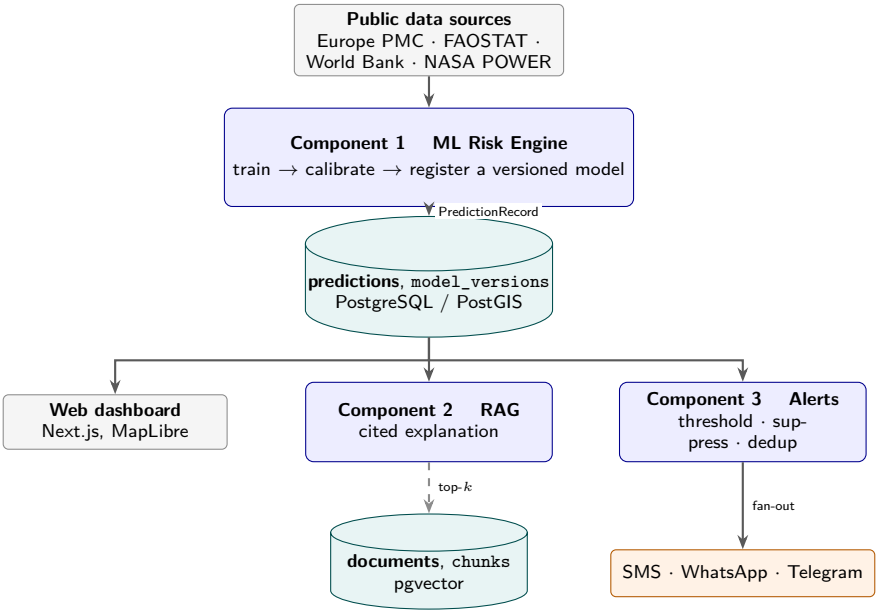


Figure 3.1: The platform decomposed into three components. The ML risk engine writes one `PredictionRecord` per area and period; the dashboard, the RAG layer and the alert service consume it independently. No component reads another component's tables, so any one of them can be replaced without touching the other two.

2. Design Principles

Five rules govern the decomposition, and every later design decision follows from them.

One responsibility per component. The risk engine turns features into a calibrated probability, the RAG layer a prediction into a cited explanation, and the alert service a prediction into a delivered message. None computes what another could.

Contracts before implementations. Components exchange versioned schemas, never shared tables or in-process calls, so each side is built and tested against a fixture. That is what let the RAG layer be developed while the alert service was still being written.

The core is disease-agnostic. A disease is a configuration record: which label extractor, knowledge collection, templates and thresholds to use. A second disease is a configuration change plus a retrain, not a code fork.

Fail soft, never fail silent. A degraded component returns a degraded answer. If the RAG layer is unavailable the alert still goes out without its explanation; if retrieval finds nothing above the relevance floor, the layer returns a refusal plus the model's own SHAP factors. Neither case produces a wrong statement, and neither produces silence.

Every output is traceable. Each prediction carries its model version, each explanation the documents it was grounded in, each alert the prediction and rule outcomes behind it, so a decision can be reconstructed afterwards.

3. The Three Components

Table 3.1: Component responsibilities, interfaces and failure behaviour.

Component	Responsibility	Interface	On failure
ML risk engine	Features to calibrated probability, with attribution	`POST /v1/predict`, `GET /v1/risk-map`	`503` with the champion id; the last good prediction is marked stale.
RAG layer	Prediction plus question to a cited explanation	`POST /v1/explain`, `POST /v1/chat`	Alert text is sent without the explanation; chat returns a refusal.
Alert service	Prediction to a delivered, deduplicated message	`POST /v1/alerts/evaluate`	Alerts queue in Redis and drain on recovery; map is unaffected.

4. Component Contracts

A contract is a JSON schema plus its guarantees, and a narrow contract is what makes a component replaceable. The risk engine emits a `PredictionRecord` with area, period, calibrated probability, risk level, model identifier, a freshness block and the top SHAP factors. Three guarantees follow: the probability is calibrated on held-out data and reads as a frequency; the risk level is derived from it by the threshold configuration, so the two cannot contradict each other; and the factors come from the model, so they survive the RAG layer being unavailable.

The RAG layer takes an `ExplainRequest` and returns an `ExplainResponse` with an answer, resolved citations, a `grounded` flag and a refusal reason; a refusal carries no answer text. The alert service emits an `AlertEvent` with its identifier, area, period, both levels, source prediction, rule outcomes and recipient count. Components share a database but not tables, and spatial columns use PostGIS while embeddings use pgvector.

5. Modularity, Reuse and Status

The decomposition justifies its overhead only if the parts are reusable. A second disease supplies a new gazetteer, guidance collection and thresholds, while the modelling code, which never names glanders, does not change. A second LLM provider is a configuration value, because the RAG layer calls a single-method interface; a second channel is an adapter plus a template. The pipeline is a scikit-learn compatible object [32] exported with a manifest, so any Python process can load it without the API.

The report distinguishes throughout between what has been built and what has been specified: the risk engine is built and evaluated, with every number in Chapter 8 reproducible from the committed scripts, while serving and the RAG and alert layers are specified but not deployed, so no measured performance is reported for them.

Chapter 4

Methodology

1. Data Sources

Four public sources are used: a literature index, a livestock census, a development indicator set and a reanalysis climate archive [38, 35, 37, 36]. None had previously been applied to glanders risk modelling.

Europe PMC supplies the animal history: 6,674 records yielding 263 occurrence events. FAOSTAT supplies animal population, 68,449 records covering seven stock series from 1961 to 2023. The World Bank supplies socio-economic context, ten development indicators and country centroids for 217 countries from 2000 to 2024. NASA POWER supplies environment, six measures of temperature, precipitation, humidity and wind at 211 centroids from 2005 to 2024.

Extracted events rise in volume after 2015, which reflects growth in open access veterinary publishing as much as any change in incidence. This is one reason the label is treated as documented activity throughout.

2. Panel Construction

All four sources are keyed on ISO 3166 alpha-3 code and calendar year, so they join once names are reconciled. FAOSTAT is streamed from its 32 MB archive and filtered to the seven stock series rather than loaded whole. Names are matched through `pycountry` with overrides for forms such as “Türkiye”, and World Bank aggregates are excluded by requiring a three-character ISO code, so “World” never enters the panel as a country.

Combining the four sources produces a grid of 6,700 country-year rows across 268 countries for 2000 to 2024, in 60 columns: 44 engineered candidate features, three identifiers, five label fields and eight intermediate quantities.

Preprocessing reduces that grid to 2,880 rows, and selection keeps 8 of the 44 candidates, which is the matrix every model was trained and evaluated on. The row loss comes from the equine-holding filter, which keeps 162 of the 268 countries on a maximum equid stock of at least one thousand, and from the 2005 to 2023 year window; the observable-label restriction then removes nothing further. Imputation and scaling happen inside the training pipeline, so the test period cannot influence them.

The matrix is committed as `model_matrix.csv` under `data/processed/`: 2,880 rows by 8 feature columns plus the target, written by `src/11_export_model_matrix.py` from the same loader and selection the training script uses.

Events are de-duplicated on country, year and source record identifier, so one paper reporting several cases cannot contribute multiple counts. Missing values are median-imputed inside the training pipeline, with a missing-indicator feature per affected column. Missingness is low: 0.00% for FAOSTAT, 1.01% for the World Bank, 0.00% for climate, and 0.25% overall, no column above 3.8%.

3. Label Definition

The target is binary: whether a glanders occurrence was documented in country i during year $t + 1$. Occurrence is extracted from Europe PMC by matching, within one sentence, a country name from a gazetteer and a term from an occurrence lexicon (*outbreak*, *case*, *detected*, *seropositive*, *surveillance*), recorded for the year in the sentence or the publication year. Because the label depends on publication it is termed *documented activity*: a negative may mean no disease or no report, so the model predicts observable activity, not biological presence.

4. Feature Engineering

Four groups are built from information at year t or earlier. Seven disease-history features cover prior-year occurrence, a five-year total, years since the last event and activity shares. Fourteen population features cover log stocks, equid share, density, growth and species counts. Ten climate features cover temperature, precipitation, humidity and wind, each with an anomaly against the country's own expanding mean, separating unusual conditions from latitude. Eleven socio-geographic features cover income, agriculture, GDP, rural share and connectivity, and largely proxy surveillance capacity.

Three derived features need definition. Equid share is horses, asses and mules over total farm animals. Livestock units weight cattle and camels at 0.8 and small ruminants at 0.1, so one number expresses grazing pressure across species. Years since the last event uses 99 for no event in the window.

The association measures above are diagnostics: they say how each candidate relates to the label on its own, and they are not used to choose the feature set, because a marginal measure and a fitted model disagree about what matters, as Chapter 8 shows. Selection is driven by attribution instead (Section 6.).

One column is removed structurally: `equids` is exactly the sum of horses, asses and mules, which a tree can rebuild from its parts. No projection or principal component analysis is applied; with 8 retained features and 2,880 rows the binding constraint is signal rather than dimensionality.

5. Statistical Association Measures

Four measures screen the features, each from a standard library routine rather than a handwritten implementation, with versions recorded in Section 5. Let $y \in \{0, 1\}$ be the next-year label and n the number of complete pairs.

Spearman's rank correlation (`scipy.stats.spearmanr`) measures monotone association without assuming linearity, which matters because equid stock spans four orders of magnitude:

$$\rho = \frac{\operatorname{cov}\big(R(x), R(y)\big)}{\sigma_{R(x)}\,\sigma_{R(y)}}.$$

Mutual information (`sklearn.feature_selection.mutual_info_classif`, $k = 5$) captures non-monotone dependence that a rank correlation misses, estimated by the Kraskov–Stügbauer–Grassberger k-nearest-neighbour form:

$$I(X;Y) = \iint p(x,y) \log \frac{p(x,y)}{p(x)\,p(y)}\,\mathrm{d}x\,\mathrm{d}y, \qquad \hat{I} = \psi(n) - \big\langle \psi(n_x{+}1) + \psi(n_y{+}1) \big\rangle + \psi(k),$$

with ψ the digamma function and n_x, n_y per-point neighbour counts.

Cramér's V (`scipy.stats.chi2_contingency`) measures association between a categorical feature and the binary label, placing income band and region on one scale:

$$V = \sqrt{\frac{\chi^2}{n \,\min(r-1,\, c-1)}}.$$

Variance inflation (`numpy.linalg.lstsq`) quantifies how far one feature can be reconstructed from the others, and so how unstable its coefficient would be:

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 comes from regressing x_j on the remaining features. This is the measure behind the collinearity result in Chapter 8.

6. Feature Selection by SHAP

6.1 What SHAP is and why the average is over orderings

SHAP assigns each input a share of one prediction, borrowed from Shapley values in cooperative game theory. The inputs are players, the prediction is the payout, and a player's fair share is what the payout changes by, on average, across every order in which the players could have joined.

The average over orderings is the whole point. A single ordering would credit a feature with everything it adds once the features before it are already present, which depends arbitrarily on that ordering. A feature that duplicates another would look worthless if its twin entered first and essential if it entered second. Averaging over all $|F|!$ orderings removes that arbitrariness, and the weight in the formula below is simply the fraction of orderings in which the set S precedes feature j.

For feature j, with F the full set and f the model,

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \underbrace{\frac{|S|!\,\big(|F| - |S| - 1\big)!}{|F|!}}_{\text{share of orderings with } S \text{ first}} \Big[\underbrace{f\big(S \cup \{j\}\big) - f(S)}_{\text{added by } j \text{ given } S} \Big],$$

where $f(S)$ is the model output using only the features in S, the rest marginalised. Each term is the marginal contribution of j given one coalition, and the sum weights those contributions by how often that coalition occurs.

Three properties make the result usable rather than merely plausible. Local accuracy: the values sum to the prediction minus the base rate, so they decompose that prediction exactly and nothing is left unexplained. Missingness: a feature absent from the model receives zero, so the accounting does not invent contributions. Consistency: if a feature's marginal effect rises in every subset, its value cannot fall, which is the property that makes the ranking trustworthy for selection.

The implementation is `shap.TreeExplainer`, which evaluates the sum exactly for tree ensembles in polynomial rather than exponential time, so the attribution is not sampled. Global importance is the mean absolute value over the n explained rows, $\bar{\phi}_j = \frac{1}{n} \sum_i |\phi_{ij}|$, which is what ranks features for selection.

6.2 Why attribution rather than association

The four measures in Section 5. ask how a feature relates to the label on its own. Selection needs a different question: how much a feature adds to a model that already has the others. The two answers diverge sharply here. Mutual information and Spearman correlation rank `longitude` first and `temp_mean_anom` near the bottom, while attribution reverses them, because longitude is a stand-in for which continent a country sits in and the model can recover that from region, population and reporting variables. Selecting on association would therefore keep proxy columns and drop the conditional signal.

SHAP was the only attribution method used, for three reasons. It is additive and signed, so the same values supply both the global ranking that selection needs and the per-prediction top-factor list the risk engine returns, which keeps one explanation mechanism in the platform rather than two. `TreeExplainer` is exact for the tree families in use and therefore cheap enough to recompute on every training run. And it carries the consistency guarantee above, which permutation importance and impurity-based gain do not: impurity gain is biased toward high-cardinality columns, and permutation importance is unstable with correlated inputs, of which this panel has many.

6.3 Procedure and the retained set

The ranking is computed on the training years only, 2005 to 2017. Fitting on the full panel would let the held-out years influence which features were chosen, and the temporal-holdout score reported in Chapter 8 would then be optimistic. The 42 numeric candidates are ranked by mean absolute SHAP on that training window and the first 8 are kept.

8 is the whole set. No further columns are retained, and the exported dataset contains exactly those 8 columns so the artefact on disk and the model behind the results are the same object.

Table 4.1: The 8 features the reported models use, in rank order by mean absolute SHAP on the training years, with the unit each variable is measured in. The retained set carries 49% of the total attribution across all 42 candidates.

Rank	Feature	Unit as measured	Mean $\|\phi\|$
1	`pop_total`	persons	1.035
2	`years_since_last`	years (99 = none)	0.648
3	`temp_mean_anom`	°C from baseline	0.505
4	`internet_pct`	% of population	0.449
5	`humidity_anom`	% RH from baseline	0.382
6	`precip_cv`	ratio (sd/mean)	0.371
7	`log_livestock`	ln(1 + LSU)	0.364
8	`goats`	head	0.362

The set is deliberately small. 8 columns against 2,880 rows leaves an order of magnitude more rows than columns, which is what the positive count can support: with 169 events, a model given forty inputs has more directions to fit than it has events to learn from, and Chapter 8 reports what that costs in transfer.

6.4 What the ranking looks like

Panel B of Figure 8.3 plots the mean absolute SHAP value against the marginal rank for the full candidate set, and the absence of any downward trend is the disagreement described above made visible. The eight retained features occupy the top of the conditional ordering; features that lead the marginal ordering sit far below it.

7. Models

Six families are compared. Logistic regression is the additive baseline with balanced class weights. A single shallow decision tree is included because a non-specialist can read it. Random forest and extra trees are bagged ensembles [20], differing in split randomisation; XGBoost and LightGBM are boosted ensembles [18, 19]. Tuning used grouped cross-validation on country. Neural networks were excluded: with 169 positives and 8 features a network has more parameters than examples and would memorise the training period.

8. Evaluation Design

Three schemes answer different questions. A temporal holdout trains on 2005 to 2017 and tests on 2018 to 2023, with the test period never observed during fitting, which answers whether the model can be used in following years. Leave-country-out uses five folds grouped by country, so every prediction is for an unseen country; it answers whether the model transfers and it governs selection, being the harsher test and the closer approximation of deployment. Repeated stratified cross-validation, five folds three times, is a stability check.

Precision-recall AUC is the primary metric: at 5.87% prevalence ROC-AUC is dominated by the true-negative mass and overstates performance [22, 23], so it is reported only for comparability. Brier score and expected calibration error are reported because an uncalibrated probability cannot support a threshold-based alert [25, 26]. Precision, recall, specificity, F1 and the Matthews correlation use a threshold tuned on training data only, with bootstrap percentile intervals over one thousand resamples. Grouped rather than random splitting is essential, since a random row split would let the model memorise country identity [24].

9. Leakage Controls

Six controls apply: features use year t or earlier and only the target uses $t+1$; the split is chronological; hyperparameters are tuned inside training folds; calibration is fitted on training folds only; the alert threshold is fixed on training; and imputation values and missing indicators are generated inside the pipeline.

Chapter 5

MLOps and Model Serving

1. From a Notebook to a Service

A trained model is not a component until it can be called, versioned, monitored and replaced. Three things must be true first. The model must be a portable artefact, so the object scored in evaluation is the object served, with nothing re-fitted in between. Every prediction must name the version that produced it. And the service must degrade predictably, so a bad release or a missing feature does not produce confident nonsense. The project is a sequence of numbered scripts, each producing artefacts consumed by the next, with every intermediate output cached so any stage can be re-run alone. That is also what makes the pipeline inspectable: a reader can run one stage and check it before trusting the next.

2. The Model Bundle

The unit of deployment is a directory, not a pickled object, because a pickle alone cannot say what its inputs mean.

```
gw-xgb-2026.11.1/
  model.joblib        # fitted pipeline, predict_proba
  manifest.json       # feature order, dtypes, calibration, metrics
  thresholds.json     # level boundaries and hysteresis margin
  SHA256SUMS
```

The manifest is what makes the bundle reusable. It records the ordered feature list with dtypes, so serving rejects a request with the wrong columns before any inference happens. It records the calibration method and achieved error, so a consumer knows whether to read the output as a probability or only as a rank. It records the evaluation metrics, so the number in the report and the number in the registry are the same. Because the bundle is a directory with a manifest, the API, a batch job and a notebook load it with identical behaviour.

Every training run also writes a record with the git commit, data version, feature list and hash, hyperparameters and the full metric set for all three schemes. Promotion is decided on leave-country-out precision-recall AUC rather than on the more flattering temporal holdout. A candidate becomes champion only if it beats the incumbent by more than the bootstrap interval width, its expected calibration error is below the deployment ceiling of 0.05, its feature schema matches what serving can supply when checked by name and order, and it has scored a shadow period without a prediction outside the plausible range. The previous champion is retained, so promotion is reversible, and a bundle failing its checksum or schema check is replaced by the previous champion rather than served.

3. Serving Design

The interface is deliberately coarse: a caller asks for a prediction for an area and period and receives the full record, rather than assembling one from several calls.

Table 5.1: HTTP interface of the risk engine.

Endpoint	Method	Purpose and contract
`/v1/predict`	POST	Score one area and period from a supplied or stored feature vector. Returns one `PredictionRecord`.
`/v1/predict/batch`	POST	Score up to 500 area-period pairs, with per-item errors rather than failing the request.
`/v1/risk-map`	GET	GeoJSON of the latest predictions for a period, filtered by risk level. Serves the dashboard directly.
`/v1/models`	GET	Registry listing: model id, promotion date, metrics, feature-schema hash.
`/v1/models/{id}`	GET	The manifest of one version, for inspection by a consumer.
`/v1/health`	GET	Liveness and readiness, where readiness checks that a champion bundle is loaded and its schema resolves.

Every response carries the `model_id` that produced it and a `data_freshness` block. The alert service reads that block before sending anything, so a prediction computed on stale upstream data cannot produce a message.

The order of operations is a reliability choice. Validation runs first, so a malformed request costs a schema check rather than a model load. The cache is consulted before inference because risk maps are read far more often than recomputed. Inference is deterministic for a fixed bundle, so a cache key of model id, area and period is safe, and the model id is part of the key rather than a version counter so promotion invalidates the cache. Responses are written back to the predictions table as well as cached, so the map does not depend on a warm cache.

The endpoint is specified against a budget rather than measured, since the service is not deployed: 40 ms on a cache miss and 4 ms on a hit, of which feature retrieval takes 25 ms and inference 8 ms. The explanation endpoint has a separate budget of two seconds because it includes generation, which is why the alert service never calls it inline. A message must not wait on an LLM.

4. Monitoring and Retraining

Three signals are monitored. Service health covers request rate, error rate and latency percentiles per endpoint on Prometheus and Grafana, catching the least interesting failure. Data drift compares each feature's current distribution with the training one using a population stability index and a Kolmogorov-Smirnov statistic, since a tree extrapolating outside its training range returns a confident answer with no basis. Calibration decay tracks the mean predicted probability against the observed event rate: a model that keeps saying 0.2 while the observed rate falls to 0.05 has lost calibration even with its ranking intact.

Retraining follows from the third signal. Recalibrate when the observed-to-predicted ratio leaves a band for two consecutive periods, and retrain when a new year of labels is available and the champion's leave-country-out score has degraded beyond the bootstrap interval.

The pipeline runs linting, unit tests and a smoke test on every push. Unit tests cover what can be checked without data: schema validation, the level-threshold function including hysteresis at the boundary, idempotency key derivation, and the citation parser. A further test loads the committed bundle, scores a fixed fixture and asserts the output to four decimals, which catches the most common silent failure here, where a dependency upgrade changes numerical behaviour and every downstream number moves unnoticed.

The full computation runs in about 4.2 minutes on an Apple M4 processor with 10 cores and 16 GB of unified memory without a graphics accelerator. Per-family fitting time at the country-year level ranges from 1.3 seconds for the decision tree to 64.2 seconds for LightGBM. One module-level seed governs fitting, cross-validation, bootstrap resampling and SHAP sampling, with software versions Python 3.12.13, scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0, SHAP 0.54.1, NumPy 2.5.4, pandas 3.1.1, SciPy 1.18.1, and the bundle smoke test runs single-threaded because tree ensembles are not bitwise reproducible across thread counts.

Chapter 6

The RAG Explanation Layer

1. The Problem This Layer Solves

A calibrated probability of 0.24 is actionable for a veterinary officer and meaningless to an equine owner. The owner needs to know what happened and what to do next; the officer needs the factors behind the number, because a forecast they cannot interrogate is one they will not act on.

Asking a large language model is the obvious answer and the obvious hazard. A model asked about glanders answers from training data that may be outdated, may conflate glanders with melioidosis, and may invent a treatment protocol; for a zoonosis under statutory control, an invented instruction is worse than none. Retrieval-augmented generation [27] changes what the model may use: it is given a small set of passages from official guidance plus the prediction's own numbers, and required to answer only from those, so each claim traces to a passage the reader can see.

The goal is narrow and testable: an answer a veterinary officer would accept as consistent with the National Action Plan, and a refusal rather than a guess when the corpus does not support one.

Fine-tuning would produce fluent text with no citations, no way to separate guidance from the base model's priors, and knowledge frozen at training time. Retrieval keeps knowledge in a corpus a re-ingestion can update, makes every answer auditable, and allows a refusal path, at the cost of latency and possible retrieval failure.

2. Knowledge Base

The corpus is restricted to authoritative sources, because the layer's value is that it cites them. Each document carries an authority tier, used as a tie-breaker during retrieval.

Table 6.1: Knowledge base, by authority tier.

Source	Tier	Use
DAHD National Action Plan for Glanders, 2025 [34]	1	Case definitions, surveillance obligations, movement control, quarantine, notification duties.
WOAH and WHO disease information sheets	1	Disease biology, transmission, zoonotic risk, official status.
ICAR and national laboratory guidance	2	Diagnostic sampling, laboratory referral.
Open-access publications	3	Epidemiology, risk factors, regional situation.

Tier 3 may inform an explanation but never overrides tier 1 on a control instruction. Conflicts resolve to the higher tier and are logged, because a silent contradiction between sources is the failure this layer exists to prevent.

3. Ingestion, Embedding and Retrieval

Ingestion runs offline as a job and stores each chunk with enough metadata to be cited and filtered. Chunking is section-aware rather than fixed-width, because a fixed window cuts control instructions in half and lets the model complete a truncated sentence. The splitter respects headings first, then paragraphs, falling back to a sliding window only past the ceiling. Chunks target 400 to 700 tokens with 15 per cent overlap inside a section, and sections under 80 tokens merge into the parent. The full section path is retained, for example `NAP-2025 > 5.2 Movement control`, so a citation points to a subsection rather than a 90-page document.

Chunks are embedded with a compact sentence-transformer [28] producing 384-dimensional vectors, with a multilingual variant for Hindi because an English-only embedding degrades on Devanagari. Vectors live in pgvector beside the relational tables, so re-ingestion and prediction share one transaction boundary. The index is HNSW with cosine distance [29]; the reranker corrects recall at the head of the list, and the index keeps query cost flat as the corpus grows.

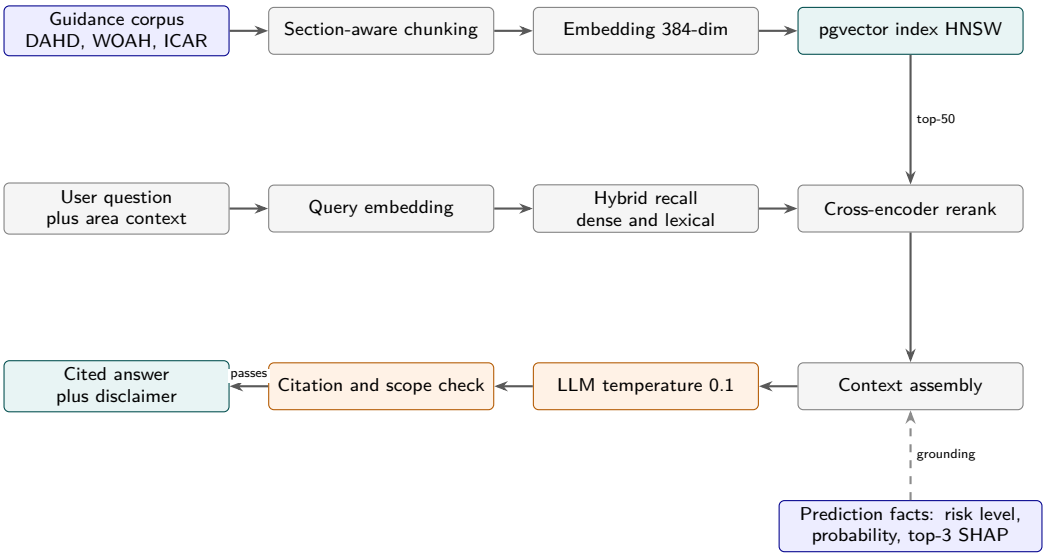


Figure 6.1: The RAG layer. Ingestion (top row) is an offline job; retrieval, assembly and generation (below) run per request. Predicted risk and SHAP factors are injected as structured facts, so the answer explains this forecast rather than glanders in general. The citation check sits after generation and can return a refusal instead of an unsupported answer.

Retrieval is hybrid, because dense and lexical search [30] fail differently here. A dense query for "movement control" may miss a passage using "restriction of equid movement", where a lexical match succeeds; conversely a lexical query for a rare acronym retrieves noise a dense query ignores. Both lists are fused by reciprocal rank fusion [31], which combines rankings without requiring comparable scores:

$$\mathrm{RRF}(d) = \sum_{i \in \{\text{dense},\text{lexical}\}} \frac{1}{k + \mathrm{rank}_i(d)}, \qquad k = 60$$

The fused top fifty are reranked by a cross-encoder scoring each passage against the question jointly. It is far slower per pair than an embedding dot product, which is why it runs on fifty candidates rather than the corpus, and the top five become the context.

Table 6.2: Retrieval parameters.

Stage	Setting	Reason
Dense recall	top 30, cosine	Semantic paraphrase coverage
Lexical recall	top 30, BM25	Exact terms, acronyms, section titles
Fusion	reciprocal rank, $k = 60$	Scale-free combination of two rankings
Rerank	cross-encoder, top 50 to 5	Precision at the head of the list
Relevance floor	rerank score below 0.30 discarded	Prevents padding the context with weak passages

The relevance floor drives the refusal path: if fewer than two passages clear it, the layer returns a refusal with the SHAP factors, because marginal passages produce a confident answer built on nothing.

4. Context, Generation and Guardrails

The prompt has three parts in fixed order: a system instruction, the retrieved passages with identifiers, and the structured prediction facts. Those facts make the answer specific to this forecast rather than to glanders in general, and are injected as structured data so the model cannot restate an association as a cause. The instruction states four constraints: answer only from the passages, cite every claim, never diagnose an individual animal, and state that the output is a population-level forecast.

Generation uses temperature 0.1 and a bounded output length, chosen for consistency rather than caution: two officers asking the same question about the same area should receive the same instruction. The provider sits behind a single-method interface, so one retrieval, grounding and citation path serves a local Ollama model or a hosted API, and switching model cannot silently disable the guardrails.

The layer is designed to refuse, and four checks can terminate a response.

Table 6.3: Guardrails and their failure behaviour.

Check	What it verifies	On failure
Retrieval floor	At least two passages above the relevance threshold	Refusal with the top SHAP factors, which need no retrieval
Citation resolution	Every cited identifier exists in the retrieved set	Answer rejected and regenerated once, then refusal
Diagnosis scope	No individual-animal diagnosis or treatment prescription	Response rewritten without the clinical claim, and the attempt logged
Out-of-scope	Question is about glanders control, not unrelated topics	Refusal stating what the layer covers

The citation check is mechanical: the answer is parsed for identifiers and each is looked up in the retrieved set, and one that does not resolve means the model cited a document it was not given, which is a hallucination whether or not the sentence is true. Asking a model to be accurate is not a control; checking its output is.

5. Evaluation

Five quantities are defined below, following the project proposal. None is reported as a measurement, because the layer has not been run against the corpus; measuring them is future work (Chapter 9).

Retrieval quality. Recall@k, the fraction of hand-built questions whose known relevant passage appears in the top k results, at $k = 5$, with mean reciprocal rank, which rewards returning it near the top rather than merely inside the cut-off. This isolates retrieval from generation, since no generation quality helps if the right passage is never retrieved.

Faithfulness. The fraction of claims the retrieved context supports, scored by a judge model and reviewed by hand on a sample; the RAGAS-style metric that catches a model citing a passage while saying something the passage does not say.

Answer relevance. Whether the answer addresses the question asked rather than reciting correct but tangential guidance, scored on the same sample, since an answer can be grounded and still unhelpful.

Citation precision. The fraction of citations resolving to a passage that supports the sentence, distinct from resolution, which only checks existence.

Refusal behaviour. The refusal rate on out-of-scope questions and the false refusal rate in scope: a layer that never refuses is unsafe and one that always refuses is useless.

Deployment targets, so the claim is falsifiable: recall@5 above 0.85, faithfulness and answer relevance above 0.90 and 0.85, citation precision above 0.95, a false refusal rate below 10 per cent, and p95 latency under two seconds.

Chapter 7

Alerting System

1. Purpose and Scope

The GlandersWatch alerting system converts a machine-learning risk prediction into a controlled notification for registered recipients. The risk engine estimates the probability of documented Glanders occurrence for an area and period; the alert service decides whether to notify, who should receive the message and how delivery is managed. A prediction is not itself an alert: every new prediction is evaluated, but a notification is generated only when the configured policy requires it.

The design is period-aware and uses one workflow for monthly and yearly predictions. Monthly prediction is future work for the ML component; the currently evaluated model is country-year based. The alert service is designed to accept monthly records once a monthly model, target and data pipeline have been developed and validated. Its output is a population-level warning, not an individual-animal diagnosis or a replacement for veterinary examination and laboratory testing.

2. Related Alerting and Warning Systems

The reviewed systems address different parts of warning and alert management. They provide useful design lessons, but none should be presented as implementing the complete GlandersWatch workflow.

2.1 NADRES v2

NADRES v2 is associated with India's animal-disease reporting and surveillance activities. It highlights the importance of organised disease information and institutional reporting. GlandersWatch uses this surveillance context as background, but adds a predictive risk input and a separate process for deciding whether recipients should be notified.

2.2 Google and Central Water Commission Flood Warnings

Public flood-warning services demonstrate how hazard information can be communicated to affected people in a timely and understandable way. The lesson for GlandersWatch is to communicate the affected area, risk level and practical action, rather than expose only a probability. Flood-specific thresholds and procedures cannot simply be transferred to animal-disease warnings.

2.3 J-PAL Bihar Flood-Warning Study

The J-PAL Bihar work is relevant because it considers whether early-warning information reaches people and supports preparedness. It suggests that warning effectiveness should be assessed beyond message generation: whether people receive, understand and act on a warning also matters. GlandersWatch can use this evaluation perspective, while recognising that animal-disease response differs from flood preparedness.

2.4 Prometheus Alertmanager

Prometheus Alertmanager provides software-alert concepts such as grouping, inhibition, silencing and routing to reduce repeated notifications during an incident. GlandersWatch adapts the general idea of separating alert conditions from notification routing. It additionally needs disease-risk periods, recipient subscriptions, veterinary guidance and clear forecast disclaimers.

2.5 OASIS Common Alerting Protocol (CAP)

CAP is a standard format for exchanging public-warning information between systems. Its relevance is interoperability: external systems can consume structured alert information without depending on the sender's internal implementation. CAP may be a future export/integration format; it is not part of the internal risk-classification rules.

2.6 Lessons and Comparison

Table 7.1: Previous systems and the lessons applied in GlandersWatch.

Previous work	Main focus	Lesson for GlandersWatch
NADRES v2	Animal-disease surveillance and reporting.	Use structured disease information and preserve traceability.
Google/CWC warnings	Communication of hazard information to affected people.	Provide understandable, action-oriented messages.
J-PAL Bihar study	Reach and usefulness of early warnings.	Evaluate receipt, understanding and response, not only sending.
Prometheus Alertmanager	Grouping, silencing, inhibition and routing.	Control repeated notifications and separate alerting from delivery.
OASIS CAP	Standardised public-warning exchange.	Keep alert information structured for possible external integration.

The proposed contribution is the integration and adaptation of these ideas around GlandersWatch's calibrated area-period disease-risk prediction. The design combines prediction-aware risk-state transitions, configurable suppression, three-level duplicate protection, area-based subscriptions, asynchronous delivery and an audit trail. This is not a claim that the individual mechanisms are new or that every previous system lacks them; the systems solve different problems, while GlandersWatch organises the relevant mechanisms for its animal-disease use case.

3. Proposed End-to-End Architecture

Figure 7.1 shows the supplied end-to-end architecture. The flow begins with user registration and prediction generation, then proceeds through evaluation, rules, alert creation, fan-out and notification. The data layer stores predictions, subscriptions, period-specific alert state, alerts and delivery logs. The approval gate is optional and may hold selected HIGH alerts for veterinary review; its use requires agreement with project and veterinary stakeholders.

3.1 Prediction Trigger and Alert Evaluator

After the risk engine stores a prediction, it publishes `predictions_ready`. The event tells the evaluator that a prediction batch is available; it does not mean a message must be sent. The evaluator loads the new prediction, previous state for the same area and frequency, active subscriptions and applicable configuration. It passes these inputs to the rules core. It does not train the model or calculate the probability.

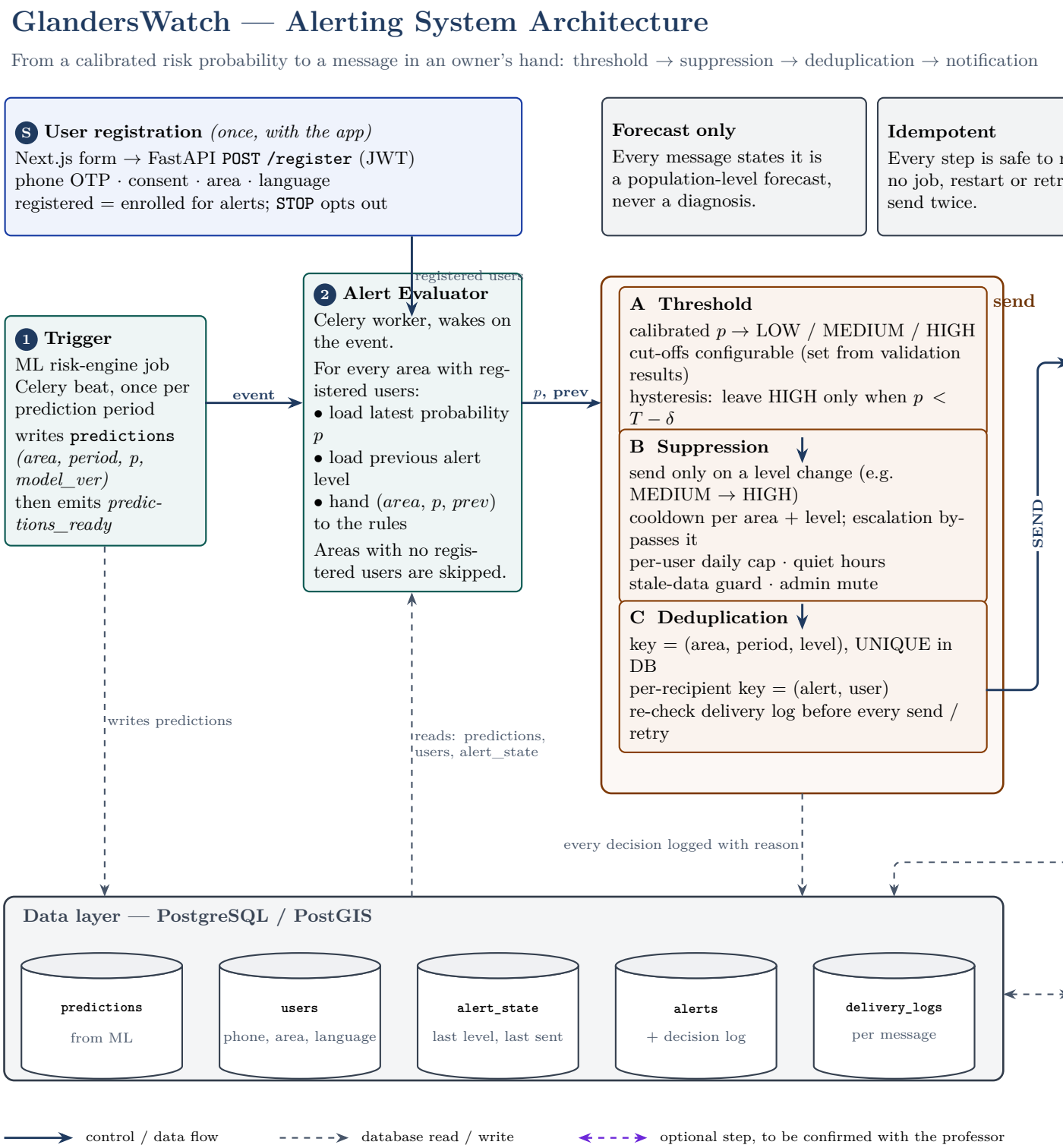


Figure 7.1: End-to-end GlandersWatch alert pipeline. The period field supports monthly or yearly predictions.

Table 7.2: Difference between reviewed work and the proposed design.

Aspect	Reviewed work provides	GlandersWatch proposes
Input	Surveillance reports, hazard information, monitoring conditions or prepared alerts.	Calibrated Glanders risk prediction with area, period, model version and freshness.
Period	Frequency depends on each source system.	Common period-aware contract for monthly and yearly predictions.
Risk state	Individual warning or alert-management capabilities.	LOW/MEDIUM/HIGH classification, hysteresis and previous-state comparison.
Notification control	Examples include routing, silencing, grouping or warning distribution.	Combined unchanged-level, cooldown, escalation, stale-data, mute, quiet-hour and daily-cap policy.
Duplicate control	Alert grouping or deduplication in selected systems.	Separate alert, recipient-job and send-attempt protection.
Recipients/delivery	Depends on the domain and platform.	Area subscriptions, preferences, queue-based fan-out and replaceable channel adapters.
Audit	Varies by system.	Reason-coded sent/skipped decisions and per-recipient delivery history.

3.2 Rules, Alert Record and Fan-Out

The rules engine classifies the probability, checks suppression conditions and protects alert creation against duplicates. If the decision is SEND, one logical alert record is created with its area, period, level, source prediction, reason and status. This record is separate from delivery jobs: one alert for Area A may generate 41 jobs for 40 owners and one veterinary officer.

Fan-out obtains eligible subscriptions and creates one job per subscription. Redis with Celery is a proposed production option: Redis brokers background work and Celery workers execute it. The queue is not the permanent alert-history store; PostgreSQL is proposed for durable predictions, states, alerts and delivery logs.

3.3 Notification Service, Data and Administration

For each job, the notification service checks recipient preferences and delivery restrictions, renders an approved template in the recipient's language and submits it through a channel adapter. The design allows SMS, WhatsApp and Telegram, with email or webhook adapters as extensions. Messages should state the area, period and risk level, provide approved action guidance and veterinary contact information where available, and clarify that the result is a population-level forecast, not a diagnosis.

Temporary provider failures may be retried with backoff; permanent failures such as invalid destinations are recorded without endless retries. PostgreSQL stores predictions, users/subscriptions, alert state, alert records and delivery logs. PostGIS may be used if geographic queries are needed. The admin dashboard exposes alert history, delivery status, mute controls and any approval workflow. Administrative actions should be permission-controlled and logged.

3.4 Notification Content and Recipient Action

A useful warning should be short enough to read on a phone but contain enough context for the recipient to understand why it was sent. Each message should identify the affected area, prediction period, risk category, issue date and recommended next step. It should avoid technical model terminology and should not present the probability as the chance that a particular animal is infected. The message should also identify the responsible veterinary contact or official information source, where one is available. The language and channel must follow the recipient's saved preferences and consent.

An illustrative message for a HIGH-risk transition is:

GlandersWatch risk alert – Area A
Period: March 2027
Risk level: HIGH
The model indicates an elevated area-level risk of documented Glanders occurrence for this period. This is an early-warning signal, not a diagnosis of any animal. Avoid moving or mixing animals suspected of illness, observe for unusual signs and contact your local veterinary officer for guidance. Do not handle suspected cases without appropriate advice.
For assessment and reporting, contact: [local veterinary contact].

The exact action wording should be approved by veterinary authorities before deployment and adapted to local language and operating procedures. A message should not disclose another user's contact details or unnecessary personal information. If no verified local contact is configured, the system should direct recipients to an approved general veterinary service rather than inventing a phone number. Message templates should be versioned so the content used for each alert can be reconstructed during an audit.

4. Period-Aware Prediction Contract

The prediction must identify its frequency and exact period. A conceptual monthly record is shown below; field names are illustrative and can be aligned with the final API schema.

```
{
  "area_id": "AREA-A", "period_type": "month",
  "period_id": "2027-03",
  "period_start": "2027-03-01", "period_end": "2027-03-31",
  "probability": 0.34, "model_version": "gw-model-v1.2",
  "data_freshness": {"passes": true}
}
```

For yearly records, `period_type` is `year` and the period ID identifies the year. The same evaluator and rules core process either type. State is stored separately by area and frequency: a March 2027 monthly result must not overwrite Area A's 2027 yearly state.

A period-aware logical alert key is:

$$K_{alert} = (\text{area_id}, \text{period_type}, \text{period_id}, \text{risk_level}).$$

This prevents a monthly alert from colliding with a yearly alert for the same area. Cooldowns, freshness requirements and thresholds may also be configured by frequency; monthly settings should not simply inherit yearly values without validation.

Table 7.3: Monthly and yearly operation in the same alert service.

Aspect	Monthly	Yearly
Period identity	Area + month ID	Area + year ID
Trigger	Monthly prediction-ready event	Yearly prediction-ready event
Previous state	Monthly stream state	Yearly stream state
Deduplication	Area + month + level	Area + year + level
Policy	Monthly thresholds/cooldown as validated	Yearly thresholds/cooldown as validated

5. Rules Engine: Thresholding, Suppression and Deduplication

5.1 Thresholding and Hysteresis

The calibrated probability is mapped to LOW, MEDIUM or HIGH. Numerical boundaries must be selected from model validation and the trade-off between false alerts and missed events; 0.5 must not be assumed to be the correct HIGH threshold.

Table 7.4: Meaning of the proposed alert levels.

Level	Interpretation	Example guidance
LOW	Below configured MEDIUM boundary.	Routine monitoring; normally no risk-change message.
MEDIUM	Between MEDIUM and HIGH boundaries.	Monitor animals, prepare and keep veterinary contacts available.
HIGH	At or above configured HIGH boundary.	Follow approved precautions, avoid unnecessary movement of suspected animals and contact a veterinarian/authority.

Hysteresis reduces frequent switching near a boundary. An area enters HIGH when $p \geq T_H$ and leaves HIGH only when $p < T_H - \delta$, where T_H is the HIGH entry threshold and δ is a configurable margin. For illustration, if $T_H = 0.30$ and $\delta = 0.05$, a probability of 0.32 enters HIGH; a later value of 0.29 remains HIGH; the area leaves only below 0.25. The margin must be validated for the relevant prediction frequency.

5.2 Suppression

Suppression decides whether a notification is necessary; it does not delete or ignore the prediction. The evaluator still processes each new prediction. The rules are:

1. **Unchanged level:** normally notify on a risk-level transition, not every period. MEDIUM in consecutive months does not automatically create repeated messages.
2. **Cooldown:** hold a repeat of the configured type/level within a time window; for example, a 48-hour cooldown can hold a repeat inside that period.
3. **Escalation bypass:** a permitted upward change, such as MEDIUM to HIGH, can bypass ordinary cooldown so a more serious warning is not delayed.
4. **Stale-data guard:** hold or skip an ordinary alert if freshness checks fail.
5. **Area mute:** an authorised administrator may temporarily silence an area; record the action and reason.
6. **Quiet hours:** defer delivery during the recipient's local quiet period, subject to the urgency policy.
7. **Daily recipient cap:** limit messages to one recipient in a day, including when they subscribe to several areas.

Quiet hours and daily caps are checked by the dispatcher at delivery time because they depend on local time and other messages sent to that recipient. Stale data, area mute and risk-state transition are checked earlier. An optional all-clear/de-escalation message may be sent when risk falls, but it must not imply that an individual animal is disease-free.

5.3 Deduplication

Deduplication protects against replayed events, worker restarts and concurrent processing. It is distinct from suppression: suppression decides whether a message is warranted; deduplication prevents the same logical operation from being created or processed twice.

1. **Alert level:** unique key `(area, period_type, period_id, level)` prevents duplicate logical alert records.
2. **Recipient level:** unique pair `(alert_id, subscription_id)` prevents duplicate jobs for one subscription.
3. **Send level:** check delivery logs before retrying and use provider idempotency keys where available, to manage uncertain timeouts.

The first two protections should use database unique constraints, not only application checks. A local database cannot guarantee exactly-once delivery to an external provider; callbacks, status queries and reconciliation are needed where supported.

6. Modular Architecture: Core, Ports and Adapters

Figure 7.2 shows the internal modular structure. The alert engine core contains domain rules and does not directly depend on a database, web framework or messaging SDK. Four ports define what the core needs; adapters implement them for a demo or production stack.

RecipientProvider. Finds active subscriptions for an area. A demo may read a JSON file; production may query user/subscription tables or a registration service. Registration should capture verified contact details, area, language, consent and opt-out status.

RuleConfig. Supplies thresholds, hysteresis, cooldowns, daily caps, quiet hours, mute and approval settings. It may use `config.json` in a demo and database/admin UI in production. Monthly and yearly policies can differ after validation.

StateStore. Reads previous state and saves alerts, decision reasons and delivery outcomes. In-memory storage or SQLite may support a demo; PostgreSQL is proposed for durable operation. Redis may broker jobs but should not be the sole alert-history store.

Notifier. Submits messages through a common interface while hiding provider-specific APIs and authentication. Demo options include a simulated channel or JSONL outbox; possible adapters include SMS, WhatsApp, Telegram, email, webhook and CAP export. Production SMS/WhatsApp require provider onboarding and applicable template approvals.

This Ports-and-Adapters (Hexagonal) structure separates business rules from technology choices. Replacing SQLite with PostgreSQL or Telegram with an approved SMS provider changes an adapter, not the threshold and suppression logic. The core can initially run as a Python library and later be hosted behind an API; this design does not require every component to be a separate microservice.

7. Example Flow: Monthly and Yearly Predictions

The following monthly example uses fictional values only. Assume MEDIUM begins at 0.15, HIGH entry is 0.30, hysteresis is 0.05, and Area A has 40 owner subscriptions plus one veterinary officer.

Table 7.5: Illustrative monthly state sequence for Area A.

Month	Probability	New level	Previous	Decision
Jan	0.22	MEDIUM	LOW	Send transition if checks pass.
Feb	0.24	MEDIUM	MEDIUM	Suppress repeated risk-change message.
Mar	0.34	HIGH	MEDIUM	Send escalation if checks pass.
Apr	0.29	HIGH	HIGH	Remain HIGH by hysteresis; no repeated transition.
May	0.23	HIGH	HIGH	Remain HIGH; value is not below 0.25.
Jun	0.24	HIGH	HIGH	Remain HIGH; no new transition.
Jul	0.20	MEDIUM	HIGH	Optional de-escalation message.

For March, the evaluator consumes the event, loads the prediction, previous monthly state and subscriptions. The rules classify 0.34 as HIGH and detect MEDIUM-to-HIGH escalation. Freshness, mute and cooldown policies are checked; an allowed escalation bypass may override ordinary cooldown. The period-aware alert key is checked, and one alert record is created if it does not exist. Fan-out creates 41 recipient jobs, protected by the recipient unique key. The dispatcher applies quiet hours and daily caps, renders each recipient's language template, sends through the selected adapter and records each outcome.

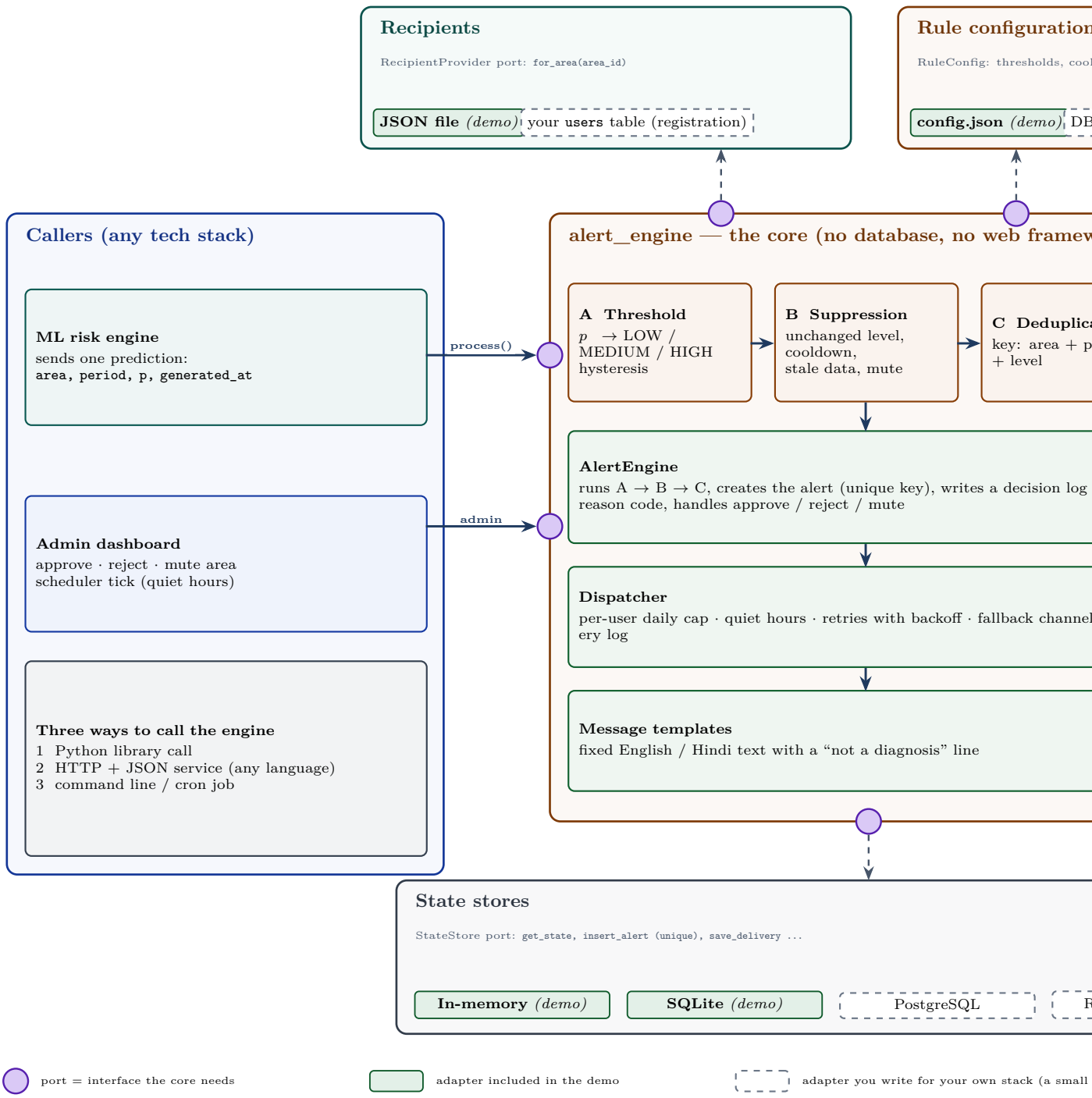


Figure 7.2: Modular alerting engine: rules core, four ports and replaceable adapters.

In April, 0.29 does not cause the area to leave HIGH because the exit boundary is below 0.25. In July, 0.20 is below that boundary and the state can de-escalate to MEDIUM. The same rules apply to yearly predictions: if the 2027 yearly state is MEDIUM and the 2028 yearly state is HIGH, the engine evaluates the transition using the yearly state and a key such as `(Area A, year, 2028, HIGH)`. Monthly and yearly states remain separate.

8. Reliability, Audit and Evaluation

Every decision, including a skipped alert, should be logged with a reason code such as `UNCHANGED_LEVEL`, `STALE_DATA`, `AREA_MUTED`, `COOLDOWN`, `DUPLICATE_ALERT` or `QUIET_HOURS`. The alert record explains why a logical alert was created; delivery logs record the outcome for each recipient.

Transient failures, such as temporary timeouts, may be retried with exponential backoff and jitter. Permanent failures, such as invalid contact details, should not be retried indefinitely. If a provider may have accepted a message but the application timed out, check the delivery log and provider status before retrying. Provider acceptance and confirmed recipient delivery should be treated as different statuses when delivery receipts are available.

Evaluation should include: (i) alert precision/false-alert rate; (ii) event coverage/missed-event rate; (iii) notification noise per recipient per month and year; (iv) delivery success by channel; (v) time from `predictions_ready` to decision and provider acceptance; and (vi) idempotency tests using event replay, worker restart, concurrent evaluation and provider-timeout simulation. These are proposed measures, not measured production results.

9. Limitations and Summary

The current evaluated model is country-year based; district-level alerts require validated district-level predictions. Monthly prediction is not yet an evaluated model output: the monthly target, input pipeline, temporal validation and calibration remain ML future work. Thresholds and hysteresis require model validation and veterinary input. Real SMS/WhatsApp delivery requires provider integration and compliance. Delivery does not prove that a recipient read or acted on a message. Finally, local database constraints prevent duplicate internal records but cannot alone guarantee exactly-once external delivery.

In summary, GlandersWatch proposes a period-aware, event-driven alert service connecting calibrated risk predictions to registered recipients. The same rules core supports monthly and yearly records through explicit period identities and separate state. Thresholding and hysteresis classify and stabilise risk; suppression controls unnecessary messages; deduplication protects alert creation, fan-out and retries; and modular ports allow storage and channel adapters to be replaced. The design adapts complementary lessons from surveillance, public warning, early-warning evaluation, alert management and standardised exchange. Monthly ML prediction, production delivery and field evaluation remain to be completed.

Chapter 8

Results and Discussion

1. The Risk Model

1.1 Dataset Characteristics

The modelling matrix holds 2,880 country-year observations, of which 169 are positive: a prevalence of 5.87% and an imbalance of 16.0 to one. 55 of 160 countries record at least one positive, 30 of them in two or more years, and the top five hold 35.5% of all positives.

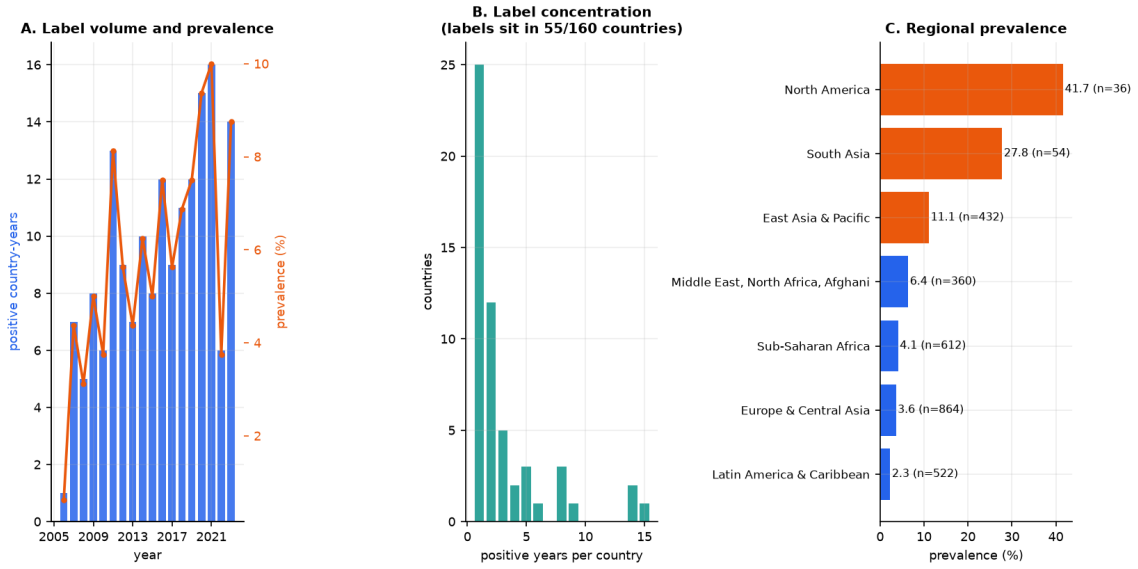


Figure 8.1: Label anatomy. Panel A gives annual positive counts against prevalence on the secondary axis. Panel B gives the distribution of positive years per country. Panel C gives prevalence by World Bank region.

Three properties shape the design. The imbalance, visible as the gap between the count and prevalence axes in panel A, rules out accuracy as a metric. The persistence in panel B, with 30 of 55 positive countries recurring, justifies historical features. The regional concentration in panel C is the first indication that the label tracks reporting capacity, since those countries operate reference laboratories.

1.2 Model Comparison

Table 8.1 reports discrimination under all three schemes, and Figure 8.2 plots the same comparison.

Two results need separating. On the forward temporal holdout the best discrimination comes from Random forest at 0.495 against a base rate of 5.87%, roughly a 8-fold lift. Under leave-country-out validation the ranking inverts: Random forest falls to 0.322, while logistic regression holds 0.381 and becomes the strongest model on the criterion that governs selection. Its temporal score is 0.461, about

Table 8.1: Discrimination at the country-year level. Brackets give 95 per cent bootstrap intervals on the temporal holdout; Brier score and expected calibration error are reported in Section 8.1.3.

Model	PR-AUC (time)	95% CI	ROC-AUC (time)	PR-AUC (country)
Logistic regression	0.461	[0.34, 0.58]	0.844	0.381
Decision tree	0.376	[0.28, 0.48]	0.871	0.243
Random forest	0.495	[0.38, 0.62]	0.874	0.322
Extra trees	0.494	[0.38, 0.60]	0.866	0.353
XGBoost	0.429	[0.32, 0.54]	0.846	0.268
LightGBM	0.471	[0.36, 0.58]	0.846	0.280

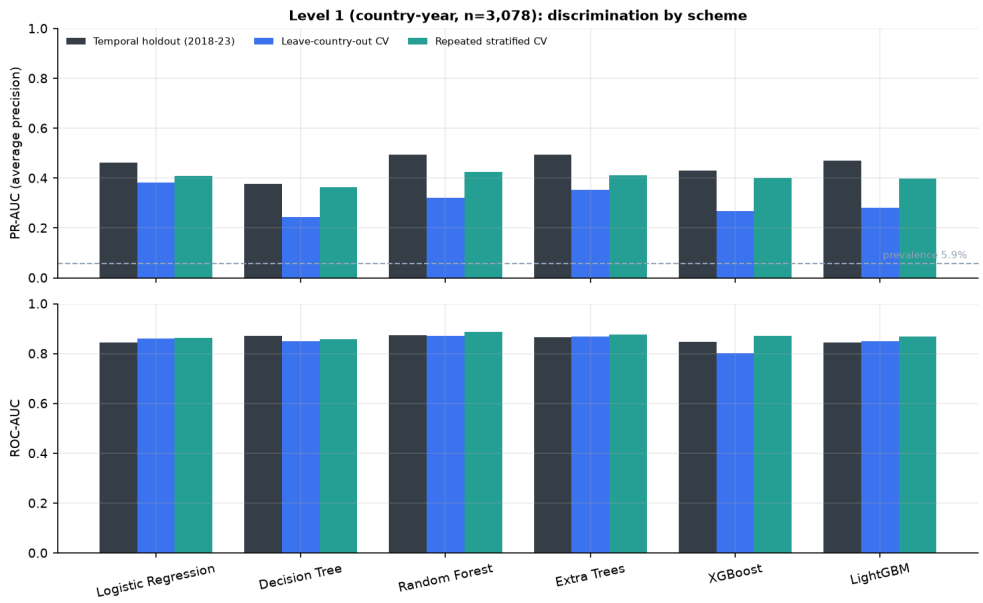


Figure 8.2: Discrimination by model and validation scheme. The gap between repeated cross-validation and leave-country-out validation is the cost of generalising to an unseen country.

a 8-fold lift.

Because deployment means an unseen country rather than an unseen year, the champion is Logistic Regression, and every number quoted below refers to it. The inversion is the more informative result. With 8 retained features the relationships are close enough to additive that a regularised linear model transfers best, and the flexible ensembles pay for their extra capacity in exactly the setting the project cares about. The single decision tree is weakest on both schemes, which confirms that the signal is not concentrated in one or two splits.

1.3 Marginal and Conditional Importance

Figure 8.3 sets the two views of feature importance against each other. Panel A plots mutual information against conditional rank, and the absence of a downward slope is the disagreement the selection section relies on; panel B gives the conditional ordering with the marginal sign annotated.

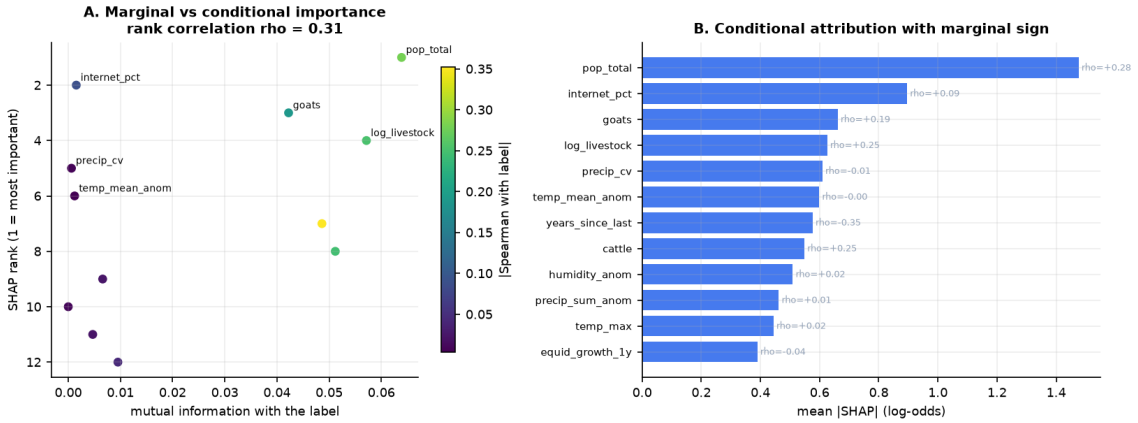


Figure 8.3: Marginal and conditional importance for the same candidate set. Panel A: mutual information against conditional rank, where a downward trend would mean the two orderings agree. Panel B: conditional rank with the marginal association sign annotated. The eight retained features are the top of the conditional ordering.

1.4 Calibration

Predictions are well calibrated, with an expected calibration error of 0.025 and a Brier score of 0.0555: a stated risk of twenty-four per cent corresponds to an observed frequency near twenty-four per cent, which is what makes a threshold-based alert defensible.

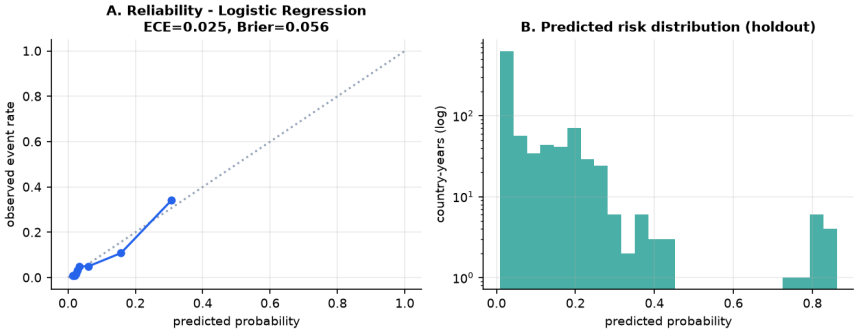


Figure 8.4: Reliability diagram for the selected model on the temporal holdout, with the distribution of predicted risk. Points on the diagonal are perfectly calibrated.

1.5 Subgroup Analysis

Table 8.2 reports leave-country-out performance by region. Skill is lowest in Sub-Saharan Africa and in low-income countries, where early warning would help most; the cause is label sparsity, since few events are recorded in those strata.

Table 8.2: Leave-country-out performance by World Bank region.

Region	Observations	Events	ROC-AUC	PR-AUC
East Asia & Pacific	432	48	0.875	0.459
Europe & Central Asia	864	31	0.824	0.161
Latin America & Caribbean	522	12	0.921	0.351
MENA, Afghanistan and Pakistan	360	23	0.825	0.422
Sub-Saharan Africa	612	25	0.738	0.112

2. The Alert Service as a Warning System

The alert layer turns the calibrated probability into messages, so its economics are reported separately from model skill.

Alerting the six per cent of country-years at highest predicted risk captures 43 per cent of observed events, half of them are reached at an alert budget of 9 per cent, and eighty per cent at 37 per cent (Figure 8.5). Uniform alerting would need full coverage for the same recall, so the model is about 5.5 times more efficient than arbitrary selection at the fifty per cent point.

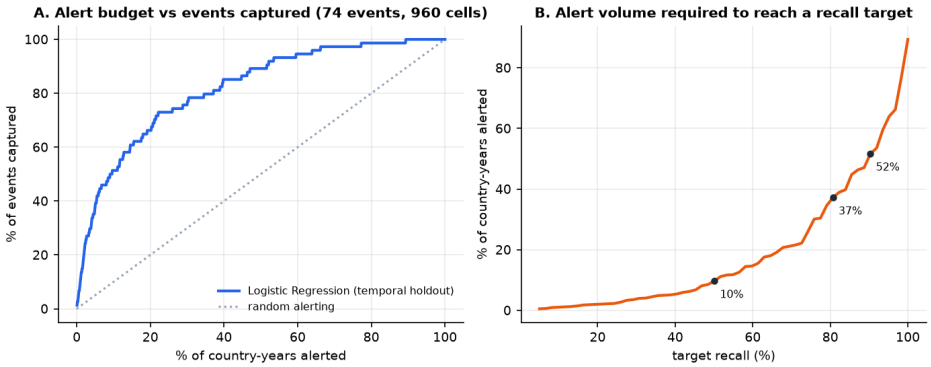


Figure 8.5: Alert economics. Panel A gives the fraction of events captured as the alert budget grows, against random alerting. Panel B gives the alert volume required to reach a target recall.

For the retrieval-augmented layer no measured results are reported: it is specified in Chapter 6 but has not been run against the corpus, so its five metrics have no values.

3. The Country-Level View

The results above answer a year-by-year question. A surveillance planner often asks a coarser one: which countries are glanders countries at all? That reframing is worth reporting because it changes the answer, and because it shows where the model's information comes from.

Collapsing the panel to one row per country, with features averaged over 2005 to 2023 and the label defined as whether the country reports glanders in any year, leaves 161 rows and 56 positives, a prevalence of 34.8 per cent rather than 5.62. Validation is leave-one-out over countries, since with one row per country there is no grouping left to hold out.

Discrimination rises to about 0.95, and the temptation is to read that as a much better model. It is not a better model; it is an easier question. Asking whether a country reports glanders ever is largely asking whether it has a veterinary reference laboratory, publishes in indexed journals and has a large

Table 8.3: Country-level classification on the same twelve features, validated leave-one-country-out. The task is easier than the country-year problem, and the scores say so.

Model	ROC-AUC	PR-AUC
Logistic regression	0.951	0.914
Random forest	0.968	0.963

equid population, and those three properties are almost perfectly separable at country granularity. The year-by-year target was hard precisely because reporting status is stable while incidence is not.

This is why the country code is not used as a feature anywhere in this report. As an identifier it cannot transfer to a country the model has not seen, which is the deployment case, and in-sample it would simply memorise the reporting capacity that the discussion identifies as the confounder. The country-level table reports the effect of that confounder rather than exploiting it.

4. Discussion

The risk model has predictive skill: Logistic Regression reaches a precision-recall AUC of 0.461 against a base rate of 5.87% on the forward split, and holds 0.381 across unseen countries, with calibration error 0.025. The gap between those two figures is smaller than the full-feature model showed, which is the practical gain from selecting features on conditional evidence. The signal it learns is surveillance rather than biology: the leading features are population, connectivity and reporting measures, the label concentrates in countries with reference laboratories, and the best-scoring cells are India, the United States and Australia.

Chapter 9

Conclusion and Future Work

1. Conclusion

This project set out to build a usable warning system for glanders: to establish whether population-level risk could be predicted from public data, and how a prediction reaches the person who owns the animals. Four sources new to the disease formed a country-year panel across 162 equine-holding countries, and six model families were trained under three validation schemes. Around the model, delivery was decomposed into three components with fixed contracts: a risk engine, an explanation layer and an alert service.

Prediction is possible, and the performance is moderate: Logistic Regression reaches a precision-recall AUC of 0.461 against a base rate of 5.87% on a forward split, about a 8-fold lift with a calibration error of 0.025, and holds 0.381 across unseen countries. Alerting six per cent of country-years captures 43 per cent of events. The skill is fragile across geography, and the decline concentrates where early warning would help most.

What the model learns is less encouraging. Three analyses converge: the highest-ranked features are population size and reporting capacity rather than biological drivers, the label concentrates in countries operating reference laboratories, and the best-scoring cells have the strongest publishing record. It predicts where glanders will be documented, not where it is.

2. Future Work

Reported activity. The most important extension is to model surveillance capacity as a quantity, as an offset on publication volume or a joint detection-and-occurrence model; otherwise gains in score reflect reporting, not disease.

The label extractor. The 263 extracted events have never been checked against a hand-annotated set; annotating a stratified sample would give the extractor's own precision and recall.

Resolution and drift. District-month resolution for India matches the scale at which control is applied, but public district covariates would need assembling. The calibration here is one point estimate, so production also needs drift monitoring and a retraining trigger.

Measuring the explanation layer. Its retrieval, guardrail and protocol targets are specified but unmeasured. Running it against the corpus would give recall@5 and mean reciprocal rank for retrieval, faithfulness and citation precision for grounding, and the out-of-scope refusal rate against the false refusal rate in scope. Two metrics belong on top: answer usefulness, rated by district officers on real predictions rather than a hand-built set, and citation age, which shows whether answers lean on guidance since revised.

Monthly alert operation. Extend the ML pipeline to produce and validate monthly predictions, then integrate them through the alert service's period-aware prediction contract. Keep monthly and yearly alert states separate, and tune thresholds, hysteresis margins and cooldown windows for each frequency. Evaluate temporal leakage, reporting delays, seasonality, calibration and freshness before enabling monthly notifications.

Production alert infrastructure. Replace demonstration storage and single-process execution with

durable PostgreSQL persistence, background workers and a reliable queue. Test event replay, concurrent evaluation, worker restarts and provider timeouts. Measure duplicate-alert prevention, queue recovery and end-to-end processing latency.

Real notification channels. Integrate approved SMS and WhatsApp providers for a consented pilot, while retaining the simulated or Telegram channel for development where appropriate. Complete provider onboarding and applicable message-template requirements. Measure provider acceptance, delivery receipts, latency, cost and failure reasons separately for each channel.

Recipient acknowledgement and escalation. Add an acknowledgement mechanism so the system can distinguish a message being sent from a warning being seen. Under a veterinary-approved policy, escalate unacknowledged HIGH alerts to the responsible veterinary officer or field worker. Evaluate acknowledgement time and follow-up completion in a supervised pilot.

Low-connectivity reach. Evaluate veterinary-officer and community field-worker relays, SMS and voice/IVR options for owners with limited smartphone or internet access. Measure whether warnings reach these recipients and whether the message is understood.

Alert quality and field evaluation. Evaluate the service separately for monthly and yearly operation using alert precision, false-alert rate, event coverage, missed-event rate, messages per recipient per month/year, delivery reliability per channel and time from alert to owner contact. Field studies should also assess whether recipients understand the warning and take appropriate veterinary or preventive action. These outcomes cannot be claimed until a real pilot with subscribers has been completed.

Structured alert exchange. If an external agency or partner requires interoperability, add an OASIS CAP export adapter for the structured alert record. Validate required fields and test exchange with a partner endpoint before claiming operational interoperability.

A WOAH WAHIS record set [33] would validate the literature-derived label where the two overlap, and a cost-sensitive threshold would match the operating point to programme priorities.

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